

Pravallika Vasantham

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SUMMARY

- Around 7 years of experience as a Python developer with analytical programming using Python and Django. Work experience in various environments such as: Python, Django, HTML/CSS, Script, Eclipse, GIT, GitHub, AWS, Linux, and Shell Scripting.
- Worked with the Relational database systems like MySQL, MSSQL, PL/SQL, Oracle, DB2, SQL Server, RDBMS, and NoSQL database systems like Redis, PostgreSQL, MongoDB, CouchDB, Cassandra, Wakanda.
- Skilled in debugging/troubleshooting issues in complex applications
- Good experience in Artificial Intelligence and Machine Learning using Python (libraries used: NumPy, SciPy, matplotlib, gensim, python - twitter, Pandas, urllib2, keras, tensorlow, nltk), CNN, RNN.
- Working with various Python Integrated Development Environments like (IDE) PyCharm, Spyder.
- Experience in Cloud (Azure, AWS, and GCP), DevOps, Configuration management, Infrastructure automation, Hadoop, Continuous Integration & Continuous Delivery (CI/CD) using bitbucket.
- Superior Troubleshooting and Technical support abilities with Migrations, Network connectivity, Security, and Database applications.
- Experience in JVM service frameworks such as Spring Boot
- Exposure to Web Application Development using Ruby on Rails, J2EE technologies and Oracle.
- Developed consumer-based features and web-applications using Python, Django, HTML
- Experience in developing web applications by following MVC Architecture using server-side application Django
- Experience in building both Data pipelines and CI/CD pipelines on premises and on cloud.
- Good knowledge in Tableau, Matplotlib, Seaborn for interactive data visualizations.
- Proficient experience with Anaplan, Power BI, Tableau within large and complex environments
- Experience in AWS Cloud Computing services, such as EC2, S3, Lambda, API, DynamoDB, EBS, VPC, ELB, Route53, Cloud Watch, Security Groups, Cloud Trail, IAM, CloudFront, Snowball, EMR and Glacier.
- Good Experience in Linux Bash scripting and following PEP Guidelines in Python
- Experience in developing ETL pipelines in and out of data warehouse using combination of Python and Snowflakes SnowSQL Writing SQL queries against Snowflake.
- Experience on Amazon Web Services for deploying using Code commit and Code deploy of EC2 instances consisting of various flavors like AWS Linux AMI, Red Hat Linux Enterprise, SUSE Linux, Ubuntu server, Microsoft Window Server2012.
- Developed entire frontend and back-end modules using Python on Django Web Framework with GIT.
- Have Experience in List Comprehensions and python inbuilt functions such as Map, Filter and Lambda.
- Experienced in Agile Methodologies, Scrum stories and sprints experience in a python-based environment, along with data analytics, data wrangling and Excel data extracts.
- Experienced in Unit testing, System Integration testing (SIT), User acceptance testing (UAT), Functional testing.
- Successfully Managed CISCO SCM team based on agile and waterfall methodologies and mission critical integrations until go live with managing complex milestones.
- Develop project deployment using Google Cloud/Jenkins, Elastic Search and using web Services like Amazon Web Services (AWS).
- Expertise in services such as Azure Kubernetes and AWS Elastic container service.
- Good exposure to UI design using Bootstrap, HTML, CSS, JavaScript. In depth knowledge and expertise in Data Structures and Algorithms, Design Patterns, proficient in UNIX Shell Scripting, python scripting and SQL Query building (SQL query with join, sub query, correlated query, and analytical query).
- Working experience in Database design, Data Modelling, Performance tuning and query optimization with large volume of multi-dimensional data.
- Performed code reviews and implemented best Pythonic programming practices.
- Hands on experience in Designing and developing API's for the applications using Express, React JS, Node JS.
- Exceptional ability to work independently with a team of Python coders.
- Self-motivated team player with good Analytical, Logical and Problem-solving ability.

TECHNICAL SKILLS

Programming Languages: Python, Ruby, C, C++, C#, Java/J2EE, JavaScript

Web Technologies: HTML5, CSS3, XML, Bootstrap3, AJAX, Dom, Springboot, jQuery, Angularjs, Nextjs Vue, Bootstrap

Databases: SQL, MySQL, Oracle, MongoDB

Frameworks: Django, Flask

Tools: & IDE: Visual Studio, Eclipse, PyTest, PyCharm, XCode

Python Libraries: NumPy, Pandas, Matplotlib

Operating systems: Windows, Linux, Mac OS

Automation tools: Jenkins, Ansible, Docker, Kubernetes, Terraform

Methodologies: Agile, Scrum, Waterfall

Cloud services: AWS S3, EC2, Boto3, Amazon EMR, Google cloud Platform

PROFESSIONAL EXPERIENCE

InfoTech Spectrum Inc,USA

September 2024-Present

Python Developer

- Developed views and templates for applications with MySQL, Python and Django's view controller and templating language to create a user-friendly interface to perform in a high-level
- Built web application by using Python, Django, AWS, J2EE, PostgreSQL, MySQL, Oracle, and MongoDB.
- Extensively worked on Data Extraction from REST API's, performed data munging, data modelling and loaded the data into Redshift DB.
- Developed backend of the application using the flask framework.
- Designed new features and made improvements to existing features in the code by fixing bugs.
- Built python scripts to read and write JSON files.
- Used NumPy and Pandas for data manipulation to check machine learning models.
- Observed the data behavior and made reusable scripts to help other teams to access for similar cases.
- Administered the application, related databases, and the hosting environments.
- Developed several REST APIs using micro-services style of architecture with Kafka as message broker and Mongo DB as backend database.
- Developed and tested many features for dashboard using React, Next, Flask, CSS, and JavaScript.
- Managed and supported Continuous Integration (CI) using Jenkins and bitbucket.
- Helped in interactive API documentation for specific Python SDK methods to write custom requirements.
- Developed framework for converting existing Power Center mappings to PySpark (Python and Spark).
- Created PySpark frame to bring data from DB2 to Amazon S3.
- Wrote Lambda functions in python for AWS's EMR, Lambda which invokes python scripts to perform various transformations and analytics on large data sets to various formats.
- Validated the developed lambda scripts and fixed the identified bugs.
- Developed microservices using AWS EMR Lambda, API Gateway, DynamoDB, RDS according to the scenario.
- Architected and detailed Position, PnL, and FICC data migration process from multiple sources into Snowflake and Vertica from Sybase - IQ, SQL Server, and Oracle.
- Coded Snowflake data loaders using Python. Reorganized large volumes of data.
- Created Python ETL pipelines for teardown and “trickle” data migration from different backends to Snowflake, SQL Server, and Vertica.
- Provided guidance to development team working on PySpark as ETL platform.
- Responsible for ETL and orchestration process using Airflow and NiFi tool.
- Added support for Amazon AWS S3 and RDS to host files and the database into Amazon Cloud.
- Debugged the application and followed messages in log files, to figure out the error if existing.
- Developed API modularizing existing python modules with the help of Pyyaml libraries.
- Utilized PyUnit, the Python unit test framework, for all Python applications.
- Creating unit test/regression test framework for working code.
- Assisted with writing effective user stories and divide the stories into SCRUM tasks.
- **Environment:** Python, Flask, AWS, Pyramid, Redis, Django, Docker, REST, GitHub, LINUX, NumPy, Node.JS, AJAX, ReactJS, Angular2, Azure Devops

University of Central Oklahoma, Edmond, OK, USA

October 2022-July,2024

IT Automation & Technical Systems Specialist

- Scripting & Automation: Developed and deployed automation scripts (Python/Bash) to streamline routine troubleshooting, system setup, and patch deployment across the campus IT infrastructure.
- Service Analytics: Conducted data-driven analysis of incident tickets using the ITSS ticketing system and SQL to identify recurring system failures and root causes, informing strategic IT service improvements.
- System Troubleshooting: Applied analytical problem-solving to rapidly troubleshoot and resolve complex hardware, software, and network connectivity issues, maintaining high user satisfaction.
- Knowledge Management: Authored and maintained comprehensive knowledge base articles, reducing support demand by enabling user self-service.

Whitehat Junior/Byju's Future School, India

September 2020 to July 2022

Python Cloud Engineer

- Developed Python based microservice to extract data from system of records into Enterprise Data warehousing.
- All micro services are written in Python utilizing distributed message passing via Kafka message broker with JSON as data exchange formats.
- Managed large-scale, geographically distributed database systems, including relational (Oracle, SQL server) and NoSQL (MongoDB, Cassandra) systems.
- Optimized Hive queries using best practices and right parameters with technologies like Python, PySpark.
- Developed spark applications in python (PySpark) on distributed environment to load huge number of CSV files with different schema into Hive ORC tables.
- Developed Templates for AWS EMR infrastructure as code using Terraform to build staging, production environments.
- Troubleshooted production issues pertaining to AWS Cloud Resources and Application Infrastructure
- Worked on reading and writing multiple data formats like JSON, on HDFS using PySpark.
- Built numerous Lambda functions using python and automated the process using the event created.
- Worked with GraphQL queries, used Apollo GraphQL library, and managed more than 75 NoSQL clusters.
- Written Infrastructure as code (IaC) in Terraform, Azure resource management, AWS Cloud formation
- Created an AWS Lambda architecture to monitor AWS S3 Buckets and triggers for processing source data.
- Created reusable Terraform modules in both GCP and AWS cloud environments.
- Analyzed hive tables using spark by integrating hive to spark.

- Led Kubernetes charts using Helm. Created reproducible builds of the Kubernetes applications, Kubernetes manifest files and releases of Helm packages.
- Developed python code to automate the ingestion of common formats such as JSON, CSV by using Logstash from elastic search to Kibana dashboard to be viewed by clients.
- Implemented a Continuous Delivery framework using Jenkins, bitbucket and Puppet in bash Linux/ UNIX environment.
- Optimizing Hadoop MapReduce code, Hive/Pig scripts for better scalability, reliability, and performance.
- Implemented and setup Continuous Integration and Continuous Delivery (CI & CD) process stack using AWS, Github/GIT, bitbucket and Jenkins.
- Implemented Automated execution of tests using ANT and Jenkins.
- Good experience in backend application development with Django, Flask, JavaScript, Next, Angular JS, MySQL
- Wrote Ansible Playbooks with Python, SSH as the Wrapper to Manage Configurations of AWS Nodes and Test Playbooks on AWS instances using Python, Ansible Scripts to provision development servers.
- Maintained a farm of AWS solution using EC2 instances, ELB's, S3, EBS, Auto Scaling and RDS. Setting up servers through using AWS for deployment or other uses for application.
- Designed roles and groups for users and resources using AWS Identity Access Management (IAM) and managed network. Used version controlling systems like GIT and SVN.
- **Environment:** AWS EC-2, GCP, Route 53, S3, SVN, Clear Case, Maven, ANT, Gradle, Jenkins, GIT, Kubernetes Web Sphere, Jira, SDLC, Docker, Nagios, Shell Scripts, Unix/ Linux bash, python, Spark, Spark API, Spark SQL

Eve19-Entertainment, Sports and Adventures

October 2019 - August 2020

Python Developer

- Developed internal auxiliary web apps using python Django framework, Ruby on Rails, with Angular, React, Next and Bootstrap CSS/HTML for reviewing sitcoms that lets users to view, add, review, up/down vote, search.
- Experienced in Git, Object-oriented programming, MySQL, and Amazon Web Services (AWS)
- Supported the development of BI portal using SQL.
- Used AWS for application deployment and configuration.
- Used NumPy to create tabular timestamp for east timestamp data manipulation and retrieval.
- Involved in various phases of Software Development Life Cycle (SDLC) such as requirements gathering, modeling, analysis, design, and development.
- Involved in tokenizing the sensitive data before archiving in AWS S3 using Enterprise Tokenization service and encrypting the data before sending it over wire to external systems.
- Built an AWS and REDIS server to store data and performed defect analysis and interaction with Business Users.
- Support customers with SDK-usage, design related queries and create high quality collateral.
- Implementation exposure on service based, SOA, RESTful technologies
- **Environment:** Python, Django, PyCharm, AWS, Git, SQL, MYSQL, RESTful, SOA, bash/LINUX, Web Sphere, TomcatWebLogic, Angular.js, React, Next, Bootstrap, NumPy

Industrial Radiographs Corporation, India

July 2018 to September 2019

Python Developer

- Responsible for the analysis, documenting the requirements and architecting the application based on J2EE standards.
- Design, and maintain develop, test, deploy the website.
- Developed entire front-end and backend modules using Python on Django Web Framework
- Designed and developed the UI of the website using HTML, AJAX, CSS, and JavaScript
- Implemented SQLAlchemy, which is a Python library for complete access over SQL.
- Designed and developed data management system using MySQL and wrote several queries to extract/store data.
- Wrote Python scripts to parse XML documents and load the data in database.
- Created real-time dashboard for Executives, utilizing Logstash, graphite, Elastic Search, Kibana & Redis.
- Generated property list for every application dynamically using Python.
- Helped with the migration from the old server to Jira database (Matching Fields) with Python scripts for transferring and verifying the information.
- Was responsible for analyzing various cross-functional, multi-platform applications systems enforcing Python best practices and provide guidance in making long term, scalable architectural design decisions.
- Used Pandas API to put the data as time series and tabular form for east timestamp data manipulation and retrieval.
- Used Test Driven Approach for developing the application and implemented the unit tests using Python Unit Test framework.
- Used PyQuery for selecting DOM elements when parsing HTML.
- Worked with JSON based REST Web services
- Deployment of the AWS web application using the Linux server with Bash scripting.
- **Environment:** Python, Django, PyUnit, PyQuery, HTML, CSS, DOM, REST, AJAX, jQuery, AWS, Git, Oracle, Linux.

Education:

- **Master of Science in Business Analytics**
 - University of Central Oklahoma, Oklahoma, USA
- **Master of Business Administration-Finance and Marketing**
 - Gitam University, India
- **Bachelor of Engineering in Computer Science**
 - Sathyabama University, India